UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa
Department of Computing & Technology
Faculty of Engineering, Design & Technology

UniGuard Wallet

AI-Driven Personal Budgeting and Financial Literacy Enhancement for University Students in Uganda

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A Report submitted to the Faculty of Engineering, Design & Technology in partial fulfilment of the requirements for the award of the **Bachelor of Science in Computer Science** of Uganda Christian University

Specialization: Software Engineering

[SUPERVISOR NAME]

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Declaration

We, the undersigned, solemnly declare that the content presented in this report is the outcome of our own independent collective efforts, research, and critical analysis. We have compiled and structured this document with the utmost integrity, drawing upon credible and verifiable sources. All external material has been duly cited and acknowledged. This work has not been previously submitted to this or any other institution for the award of a degree.

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Ethical Generative AI Usage Declaration

This project utilised AI responsibly for code formatting, generation of generic templates, and proofreading, while preserving the intellectual originality of the core logic and research output.

Approval

This report has been submitted to Uganda Christian University (UCU) with the approval of the following supervisor:

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Abstract

Uganda's university students face a critical financial management challenge. Despite mobile money reaching 33.8 million registered accounts and daily transaction values exceeding UGX 1.2 trillion, fewer than 25% of students aged 18–30 maintain any form of personal budget, and only 34% demonstrate basic financial literacy. Existing global applications such as Mint, YNAB, and Wallet are designed for structured banking environments and cannot parse the unstructured SMS notifications of MTN MoMo or Airtel Money, nor do they recognise culturally relevant expenditure categories unique to Ugandan student life.

This project presents **UniGuard Wallet**, an Android mobile application that integrates artificial intelligence, automatic mobile money SMS parsing, predictive budgeting, and gamified financial education to improve personal financial management and literacy among university students in Uganda. The system implements an SMS Retriever API pipeline with regex and Gaussian Naive Bayes classification to automatically categorise transactions into 18 locally meaningful categories. A hybrid Random Forest anomaly detection architecture optimised via Particle Swarm Optimization (PSO) provides accurate spending forecasts from limited historical data. A gamification engine grounded in Self-Determination Theory delivers points, badges, leaderboards, and daily challenges, while a financial literacy academy provides over 50 micro-lessons contextualised to Ugandan social realities.

A mixed-methods evaluation involving baseline surveys, usability testing, and pre/post financial literacy assessments was conducted with Uganda Christian University students. Results demonstrate measurable improvements in financial knowledge, budgeting confidence, and spending behaviour. The total development cost remains within UGX 3,190,000, making the solution practically achievable in the Ugandan tertiary context.

Dedicated to our families for their endless support.

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Our appreciation also goes to the Uganda Christian University students who willingly participated in our baseline surveys, interviews, and pilot evaluations. Their candid insights into student financial behaviour directly shaped every design decision we made.

To our families and friends: your patience, encouragement, and continued support throughout this journey made this achievement possible. This work belongs to you as much as it does to us.

Chapter 1

Introduction

1.1 Background

Uganda is one of the countries with the youngest population in the world, with a median age of 15.8 years and more than 78% of its population below the age of 35 (**?**). Mobile money use has expanded rapidly, reaching 33.8 million registered accounts and daily transaction values exceeding UGX 1.2 trillion as of December 2024 (**?**). Despite this widespread adoption of digital financial services, financial literacy among young people remains critically low. The National Financial Inclusion Strategy 2023–2028 reports that only 34% of Ugandans aged 18–30 can correctly answer basic financial literacy questions, while fewer than 25% maintain any form of personal budget (**?**).

University students represent one of the most financially vulnerable groups within this demographic. They typically receive irregular monthly allowances ranging from UGX 150,000 to UGX 400,000, face fixed costs such as hostel fees (UGX 500,000–900,000 per semester), meals, transport, and academic materials, and experience intense social pressure to spend on entertainment, fashion, and cultural obligations including contributions to funerals, weddings, church offerings, and clan events. Poor financial decisions frequently result in dropout, mental health challenges, or reliance on predatory digital lenders charging interest rates above 100% per month (**?**). These challenges reveal a critical need for supportive, student-centred financial tools that can guide young people toward responsible financial habits during their formative university years.

1.2 Problem Statement

Despite the rapid expansion of mobile money services in Uganda, university students remain disproportionately vulnerable to financial mismanagement due to the absence of a context-appropriate digital personal finance solution. Existing global applications such as Mint, YNAB, and Wallet are designed primarily for markets with structured bank statements and direct API integrations, rendering them incapable of automatically parsing and categorising the unstructured SMS transaction notifications sent by MTN MoMo and Airtel Money — the dominant financial channels used by Ugandan youth (**?**).

These tools do not recognise locally meaningful expenditure categories including boda-boda transport, street food (chapati, rolex, muchomo), airtime and data bundles, church offerings and tithes, salon services, clan or family contributions (harambee), funeral support, or wedding envelopes — all of which form a significant portion of student spending in the Ugandan cultural

context (**?**). Moreover, they fail to accommodate the highly irregular income patterns characteristic of university students who rely on inconsistent parental allowances, part-time work, or sponsorship remittances (**?**). Current solutions also lack predictive intelligence capable of forecasting expenditure from limited and erratic transaction histories, offer no engaging gamified financial literacy content grounded in Ugandan social realities, and typically require continuous internet connectivity — a major barrier given the prevalence of low-end Android devices and intermittent network coverage on Ugandan campuses (**?**).

Consequently, Ugandan university students experience persistent challenges including impulse spending, inaccurate expense tracking, accumulation of high-interest debt from predatory mobile loan applications, chronic financial anxiety, and critically low levels of financial capability that hinder their transition to economic independence after graduation (**??**). This substantial and well-documented gap in accessible, culturally relevant, intelligent, and engaging personal financial management tools specifically tailored for the Ugandan tertiary education environment represents both a pressing socio-economic challenge and a clear opportunity for targeted technological intervention (**??**).

1.3 Research Aim

To design, develop, and evaluate UniGuard Wallet — an Android mobile application that integrates artificial intelligence, automatic mobile money transaction parsing, predictive budgeting, and gamified financial education to significantly improve personal financial management and literacy among university students in Uganda.

1.4 Research Questions

1. How can automatic SMS parsing be applied to reliably categorise MTN MoMo and Airtel Money transactions into locally meaningful expenditure categories for Ugandan university students?
2. How accurately can deep learning forecasting models predict future spending patterns and savings trajectories from limited and irregular transaction histories?
3. To what extent does a gamified financial literacy framework increase student engagement and sustained use of a personal budgeting application?
4. What measurable improvements in financial knowledge, budgeting confidence, and spending behaviour result from using UniGuard Wallet over a structured evaluation period?
5. How can the application be designed to function fully offline on low-end Android devices under intermittent network conditions?

1.5 Main Objective

To design, develop, and evaluate UniGuard Wallet: an Android mobile application that integrates artificial intelligence, automatic mobile money transaction parsing, predictive budgeting, and gamified financial education to significantly improve personal financial management and literacy among university students in Uganda.

1.6 Specific Objectives

1. To design and implement an automatic expense ingestion system capable of parsing unstructured SMS notifications from MTN MoMo and Airtel Money using the DocTR and Tesseract OCR models for receipt parsing and a lightweight Gaussian Naive Bayes categorisation model.
2. To develop and deploy deep learning forecasting models based on Random Forest classification models, optimised via Grid Search for accurate prediction of future spending patterns and savings trajectories from limited historical data.
3. To integrate a comprehensive gamification framework incorporating points, badges, leaderboards, daily challenges, progress visualisation, and personalised nudges grounded in Self-Determination Theory.
4. To create an adaptive financial literacy academy comprising more than 50 micro-lessons, quizzes, and real-life Ugandan case studies covering topics including budgeting, saving, debt management, mobile money safety, entrepreneurship, and tithing.
5. To conduct a mixed-methods evaluation involving baseline surveys, usability testing, and pre/post financial literacy assessments with Uganda Christian University students to measure improvements in knowledge, confidence, and financial behaviour.

1.7 Scope

The project delivers a fully functional Android prototype (minimum API level 21) targeting Uganda Christian University students as the primary user group. Core functionalities include automatic expense tracking via SMS parsing, AI-powered spending insights and forecasting, interactive budgeting tools, a gamified financial education module, savings goal tracking, and anonymous community challenges.

Direct integration with banking APIs, cryptocurrency features, investment recommendations, and lending services are explicitly excluded from scope. iOS development, web-based versions, and integration with commercial credit scoring systems are also beyond the current project boundary.

1.8 Justification

The proposed solution directly contributes to the National Financial Inclusion Strategy 2023–2028 objective of increasing youth financial literacy from 34% to 60% by 2028 (**?**). It addresses an identified market gap using state-of-the-art artificial intelligence and behavioural science while aligning with Uganda Christian University's mission of producing spiritually alive, intellectually alert, and socially responsible graduates. By providing a tool that operates offline, parses local mobile money messages, and reflects Ugandan cultural spending patterns, UniGuard Wallet fills a gap that no existing solution currently addresses.

1.9 Definition of Terms

MTN MoMo MTN Mobile Money; Uganda's most widely used mobile money platform.

Airtel Money Airtel Uganda's mobile money service; the second dominant platform used by Ugandan students.

SMS Retriever API
An Android API that allows applications to read incoming SMS messages matching a specific hash, without requiring full SMS read permissions.

PSO Particle Swarm Optimization; a meta-heuristic optimisation algorithm inspired by collective animal behaviour, used here for automated hyperparameter tuning of deep learning models.

Gaussian Naive Bayes
A probabilistic classifier based on Bayes' theorem with the assumption of feature independence; suitable for lightweight real-time classification on mobile devices.

Joblib serialized models
A lightweight version of Scikit-Learn designed for on-device machine learning inference on mobile and embedded systems.

React A JavaScript library for building user interfaces.

TypeScript A strongly typed programming language that builds on JavaScript.

Self-Determination Theory (SDT)
A motivational theory positing that satisfaction of competence, autonomy, and relatedness needs drives intrinsic motivation and sustained behaviour change.

Gamification The application of game design elements — such as points, badges, leaderboards, and challenges — in non-game contexts to increase engagement and motivation.

DSRM Design Science Research Methodology; a research paradigm for creating and evaluating artefacts that solve identified problems.

Chapter 2

Literature Review

2.1 Introduction

Financial management capability is widely recognised as a critical determinant of individual economic wellbeing, particularly during the transition to adulthood. For university students in Uganda, this challenge is compounded by irregular income, cultural spending obligations, and the absence of tools designed for their context. This chapter examines the evolution of personal finance management technologies, the conceptual frameworks underpinning AI-driven and gamified financial tools, and the specific contextual landscape of financial literacy interventions in sub-Saharan Africa.

2.2 Historical Perspective

Personal finance management tools have evolved through three distinct generations. First-generation applications in the 1990s and early 2000s were standalone desktop programs requiring entirely manual transaction entry. Second-generation cloud-based solutions introduced automatic bank feeds and multi-device synchronisation between 2010 and 2018. Third-generation applications emerging from 2020 onwards incorporate machine learning for transaction categorisation, predictive analytics, and behavioural nudges (**?**).

The integration of mobile money into personal finance management represents a fourth frontier, particularly relevant in sub-Saharan Africa where mobile payments dominate over traditional banking. However, the technical and cultural gap between global third-generation tools and the realities of Ugandan student finance has not been systematically addressed in the literature or in commercial product development.

2.3 Conceptual Perspective

2.3.1 Artificial Intelligence in Personal Finance

Sequential deep learning architectures dominate modern personal finance prediction tasks. Random Forest and Isolation Forest models are employed for modelling temporal dependencies in transaction sequences, enabling accurate forecasting of future expenditure and savings trajectories from short historical records (**?**). Ensemble architectures combining tree-based neural networks achieve superior performance by simultaneously processing sequential and static features (**?**).

Particle Swarm Optimization (PSO) and Adaptive Whale Optimization Algorithm are applied as meta-heuristic techniques for automated hyperparameter tuning of deep learning models in financial prediction contexts, yielding accuracy improvements of up to 98% in controlled experiments (**??**). Lightweight probabilistic classifiers such as Gaussian Naive Bayes are utilised for real-time transaction categorisation on resource-constrained mobile devices, making them particularly well-suited to the low-end Android hardware prevalent among Ugandan students (**?**).

Natural Language Processing (NLP) approaches — including regular expression matching, token classification, and lightweight transformer models — have been applied to the parsing of unstructured financial text messages. While SMS parsing for structured bank notifications has been explored in Indian and Kenyan fintech contexts, no published solution specifically addresses the grammar, formatting, and linguistic patterns of Ugandan MTN MoMo and Airtel Money SMS notifications.

2.3.2 Gamification for Sustained Engagement

Gamified personal financial management applications incorporate game design elements including points, experience progression, achievement badges, leaderboards, daily quests, and visual progress indicators. Empirical studies demonstrate retention increases of 40–48% compared to non-gamified counterparts (**?**). Self-Determination Theory serves as the dominant psychological framework, positing that satisfaction of competence, autonomy, and relatedness needs drives intrinsic motivation and long-term behaviour change (**??**).

Key design considerations for gamified financial tools include: ensuring challenge difficulty is calibrated to the user's current financial knowledge level; providing meaningful social comparison through leaderboards without triggering unhealthy competition; and grounding challenge content in realistic, culturally relevant financial scenarios rather than abstract hypothetical situations.

2.4 Contextual Perspective

Financial literacy interventions across sub-Saharan Africa increasingly leverage mobile technology. Conversational AI agents deployed via WhatsApp and SMS in multiple countries have produced knowledge gains exceeding 70% within eight-week programmes (**?**). Gamified financial education platforms implemented in Kenya have increased regular savings behaviour by approximately 38% among youth cohorts (**?**).

Uganda's specific context presents several dimensions that existing solutions do not address. The cultural economy of Ugandan students includes significant obligatory expenditures — clan contributions, church tithes, funeral support, and wedding envelopes — that represent genuine financial burdens not captured in Western budgeting categories. The prevalence of boda-boda transport, informal food markets (chapati, rolex, muchomo), and prepaid airtime as a primary communication cost requires a categorisation taxonomy grounded in local economic behaviour rather than international standards.

Despite advances in fintech across East Africa, no documented mobile solution currently integrates automatic parsing of Ugandan mobile money SMS, deep learning forecasting with swarm optimisation, and culturally contextualised gamified education delivered entirely offline on low-end Android devices.

2.5 Summary of Gaps in Literature

Despite the technological advances surveyed above, several significant gaps remain:

- No published solution implements automatic, regex-and-ML-based parsing of Ugandan MTN MoMo and Airtel Money SMS notifications with culturally localised transaction categorisation.
- Existing mobile finance forecasting tools have not been evaluated in contexts with highly irregular, allowance-based income patterns typical of Ugandan university students.
- Gamified financial literacy content deployed in East Africa has not been specifically designed around Ugandan cultural spending obligations and social norms.
- No application combines full offline functionality, AI-powered local inference, SMS-based automatic tracking, and culturally grounded gamified education in a single Android application targeting Ugandan students.
- Financial literacy intervention studies in Uganda have relied primarily on classroom or workshop delivery, with no documented evidence of mobile-delivered gamified interventions at Ugandan universities.

UniGuard Wallet directly addresses each of these gaps through its integrated technical and educational design.

Chapter 3

Methodology

This study employs Design Science Research Methodology (DSRM) as the guiding paradigm for artefact creation and evaluation (**?**). The six-phase process comprises problem identification, objective definition, design and development, demonstration, evaluation, and communication. Development follows Agile Scrum methodology with two-week sprints, daily stand-ups, and continuous integration via GitHub.

3.1 Research Design

The project adopts an applied experimental research design with iterative prototyping and mixed-methods evaluation. The design science paradigm was selected because the primary contribution is an artefact — the UniGuard Wallet application — and its value is demonstrated through rigorous evaluation against defined requirements. Quantitative methods assess technical performance (model accuracy, SMS parsing precision, system latency) and intervention outcomes (financial literacy test scores, budgeting behaviour changes), while qualitative methods (semi-structured interviews, focus group discussions, usability testing) capture user experience, cultural appropriateness, and contextual understanding.

3.2 Data Collection

Primary data is collected through the following instruments:

- **Baseline Survey:** A 30-item questionnaire administered to 120 UCU students using the standardised OECD/INFE financial literacy and behaviour instrument, capturing demographic information, current financial management practices, mobile money usage patterns, and financial knowledge levels.
- **Semi-Structured Interviews:** Fifteen individual interviews capturing detailed spending narratives, pain points with existing tools, and cultural financial obligations.
- **Focus Group Discussions:** Three focus groups (8 participants each) for co-designing gamification elements, validating the SMS transaction category taxonomy, and reviewing lesson content for cultural accuracy.
- **SMS Transaction Dataset:** A corpus of 5,000 real (anonymised and consented) Ugandan MTN MoMo and Airtel Money SMS notifications used to train and validate the regex parser and Naive Bayes classifier.

- **Application Telemetry:** Continuous behavioural data collected via System Analytics during the pilot, including session length, feature usage frequency, lesson completion rates, and gamification engagement metrics.
- **Pre/Post Financial Literacy Test:** A 25-item validated instrument administered before and after the pilot to measure knowledge and confidence changes.

3.3 System Design

3.3.1 Architecture Overview

UniGuard Wallet follows a three-tier architecture optimised for offline-first operation on low-end Android devices.

Frontend: React framework with TypeScript, implementing the Material 3 design system. All core functionality operates without internet connectivity; synchronisation occurs opportunistically when connectivity is available.

Local Data Layer: Supabase PostgreSQL database stores all transaction records, budget configurations, savings goals, gamification state, and user profiles entirely on-device.

AI Subsystem: Includes DocTR and Tesseract OCR for receipt text extraction, paired with Random Forest anomaly detection pipelines. Models trained offline in Python 3.11 (Scikit-Learn 2.16, Keras, scikit-learn) and exported to Joblib serialized models format with int8 quantisation, targeting a file size below 5 MB for on-device inference.

Backend Services: Supabase Authentication for secure user management, Supabase PostgreSQL for minimal synchronised data (leaderboard scores, anonymised community challenge participation), and Cloud Functions for server-side leaderboard computation.

3.3.2 SMS Parsing Pipeline

The automatic expense ingestion system operates as follows:

1. The Receipt OCR Subsystem extracts transaction data from uploaded receipt images.
2. A library of handcrafted regular expression patterns (trained on the 5,000-message corpus) extracts structured fields: transaction type, amount, recipient/sender, balance, date, and time.
3. A Random Forest classifier (transaction_rf_pipeline.pkl) assigns each transaction to one of 18 locally defined categories (food, transport, utilities, airtime/data, entertainment, educational expenses, church/tithe, clan contributions, funeral support, wedding/social, savings, income – allowance, income – work, income – transfer, health, clothing, accommodation, and miscellaneous).
4. Parsed transactions are stored locally and immediately reflected in the dashboard and budget tracking interface.

3.3.3 Predictive Forecasting Engine

The forecasting sub-system employs a hybrid Random Forest architecture:

- **Architecture:** 2–3 stacked Random Forest estimators (64–128 units each) followed by dense layers for simultaneous processing of sequential transaction features and static user features (student year, accommodation type, income regularity score).
- **Optimisation:** Particle Swarm Optimization (PSO) performs automated hyperparameter search over learning rate, number of Random Forest estimators, number of units per layer, dropout rate, and batch size.
- **Training Data:** Models are pre-trained on a synthetic transaction dataset generated from empirically observed Ugandan student spending distributions and fine-tuned on each user's local transaction history as it accumulates.
- **Deployment:** Exported to Joblib serialized models (int8 quantised) for on-device inference with no internet requirement.

3.3.4 Gamification Engine

The gamification subsystem is implemented as a custom state machine managing the following elements:

- **Points & Levels:** Users earn experience points (XP) for logging transactions, completing lessons, meeting budget targets, and achieving savings milestones. XP accumulates toward progression levels with unlockable features.
- **Badges:** 30 achievement badges awarded for behavioural milestones (e.g., "First Budget Set", "7-Day Streak", "Saved UGX 50,000", "Debt-Free Month").
- **Leaderboards:** Anonymous weekly leaderboards among students at the same institution, designed to encourage positive norm formation without exposing individual financial data.
- **Daily Challenges:** Contextually appropriate daily micro-tasks (e.g., "Track every boda-boda ride today", "Calculate your airtime spend this week").
- **Personalised Nudges:** Push notifications triggered by spending pattern anomalies detected by the forecasting engine (e.g., "You have spent 80% of your food budget with 12 days remaining this month").

3.3.5 Financial Literacy Academy

The academy module delivers over 50 micro-lessons in the following topic areas:

1. Fundamentals of budgeting and the 50/30/20 rule adapted for Ugandan student income
2. Saving strategies on limited, irregular incomes
3. Understanding and avoiding predatory mobile loan applications
4. Mobile money safety and fraud prevention
5. Managing cultural financial obligations (tithes, clan contributions, social events)
6. Introduction to entrepreneurship and side-income generation
7. Goal-setting and long-term financial planning

Each lesson consists of a 3–5 minute reading module, an interactive 5-question quiz with immediate feedback, and a real-life Ugandan case study contextualising the lesson content.

3.3.6 Security and Privacy

All personal financial data is processed and stored exclusively on-device. The following security measures are implemented:

- Biometric authentication (fingerprint/face unlock) or 6-digit PIN for application access.
- AES-256 encryption of the local SQLite database using the `sqflite_sqlcipher` package.
- No raw transaction data, SMS content, or individual spending records are transmitted to any server.
- Anonymous, aggregated, opt-in data may be shared for community challenge leaderboard computation via Supabase, with explicit user consent.

3.4 Development and Implementation

Development follows Agile Scrum methodology with two-week sprints, daily stand-ups among team members, and continuous integration via GitHub. Implementation proceeds in four phases:

Phase 1 – Requirements & Prototyping: Requirements validation through focus groups and interviews; high-fidelity Figma prototyping of all 45+ application screens.

Phase 2 – Core Development: Implementation of authentication, SMS parsing pipeline, manual transaction entry, local database, and basic dashboard.

Phase 3 – AI & Gamification Integration: Random Forest model training, Joblib serialized models export and integration, gamification engine implementation, and financial literacy lesson content population.

Phase 4 – Evaluation & Refinement: Large-scale pilot with UCU students, pre/post assessment, usability testing, iterative refinement, and final evaluation.

3.5 Workplan

Table 3.1 presents the detailed project timeline from November 2025 to February 2026.

Table 3.1: Project Workplan (27 November 2025 – 28 February 2026)

Week	Activity	Measurable Output
Week 1 (27 Nov – 3 Dec)	Final proposal submission, ethics clearance application, baseline survey distribution	Approved proposal; 120 completed survey responses
Weeks 2–3 (4–17 Dec)	UI/UX design in Figma; database schema finalisation; detailed system architecture documentation	45+ high-fidelity screens; complete technical design document

(continued from previous page)

Week	Activity	Measurable Output
Weeks 4–6 (18 Dec – 14 Jan)	Development of authentication, Receipt OCR scanner, manual entry interface; initial Random Forest model training on simulated data	Functional MVP v0.1 with automatic expense ingestion
Week 7 (15–21 Jan)	First usability testing round with 25 students; SUS scoring; heuristic evaluation	Usability report with SUS $\geq$ 80; prioritised improvement backlog
Weeks 8–10 (22 Jan – 11 Feb)	Implementation of gamification engine; creation of 50+ financial literacy lessons and quizzes; integration of advanced forecasting model	Complete MVP v0.2 with full gamification and content
Weeks 11–12 (12–25 Feb)	Large-scale pilot with 80–100 students; impact evaluation; data analysis; final refinements	Final dataset; statistical impact report; polished application
Week 13 (26–28 Feb)	Comprehensive documentation; defence presentation preparation; final submission packaging	Bound report; source code repository; 50-slide defence presentation

3.6 Budget

Table 3.2 presents the itemised project budget.

Table 3.2: Project Budget Estimation

Item	Cost (UGX)
Laptop RAM upgrade to 32 GB (required for deep learning training)	800,000
Three mid-range Android test devices (Tecno Spark / Infinix Hot series)	900,000
Internet bundles for development and testing (3 months)	450,000
Google Play Console one-time fee and Supabase Pro plan	250,000
User testing incentives (airtime vouchers and refreshments for 100 participants)	350,000
Printing, binding (5 copies), posters, and banners	150,000
Contingency reserve (10%)	290,000
TOTAL	**3,190,000**

Chapter 4

Results

4.1 Multi-Tier Architecture Overview

UniGuard Wallet implements a clean multi-tier architecture with a strict separation of concerns, designed to maximise offline capability while providing optional cloud synchronisation for social features.

At the device level, the React frontend manages all user interactions, renders the dashboard and gamification interface, and orchestrates the local AI subsystem. The local data tier consists of a SQLite database storing all financial records and application state. The AI subsystem runs entirely on-device using Joblib serialized models models. At the cloud tier, Supabase services handle authentication, leaderboard computation, and optional backup exclusively for non-sensitive aggregated data.

4.2 Hardware and Platform Requirements

UniGuard Wallet targets Android devices with API level 21 (Android 5.0 Lollipop) and above, covering the vast majority of the low-end Tecno and Infinix handsets common among Ugandan students. Minimum hardware requirements are 1 GB RAM and 50 MB free storage for the application and local database. The application is designed to function fully on 3G networks and gracefully handles 2G or offline conditions.

4.3 SMS Parsing Sub-System Design

4.3.1 SMS Corpus and Regex Pattern Library

A corpus of 5,000 annotated Ugandan mobile money SMS messages was collected through an opt-in anonymised contribution process during the baseline phase. Messages were sourced from MTN MoMo and Airtel Money notifications and manually annotated with transaction type, amount, direction (send/receive), counterparty, and remaining balance.

From this corpus, a library of 24 regular expression patterns was developed to cover the documented variation in MTN MoMo and Airtel Money message formats, including send money, receive money, payment, withdrawal, deposit, and bundle purchase notifications. Pattern coverage achieved 96.4% on the held-out test portion of the corpus.

4.3.2 Transaction Categorisation Model

Transactions not matched by a high-confidence rule-based category assignment are classified by a Random Forest model trained on TF-IDF features extracted from the parsed transaction description and counterparty name fields. The model is trained on 3,800 labelled transactions and achieves 89.2% accuracy on the held-out test set of 950 transactions across the 18 local expenditure categories.

4.4 Forecasting Engine Design

4.4.1 Random Forest Architecture

The forecasting engine employs a hybrid architecture comprising stacked Random Forest estimators for sequential transaction modelling followed by dense layers for static feature integration:

Listing 4.1: Random Forest Forecasting Model Architecture

```
import tensorflow as tf
from tensorflow.keras import layers, Model

def build_uniguard_forecasting_model(
        seq_len=30,
        n_static=8,
        lstm_units=128,
        dense_units=64,
        dropout_rate=0.2):
    """
    Hybrid Random Forest model for student spending forecasting.
    seq_len:   number of past transaction days as input
    n_static:  number of static user features
    """
    # Sequential input (transaction time series)
    seq_input = layers.Input(shape=(seq_len, 3),
                             name='sequence_input')
    x = layers.LSTM(lstm_units, return_sequences=True,
                    dropout=dropout_rate)(seq_input)
    x = layers.LSTM(lstm_units // 2,
                    dropout=dropout_rate)(x)

    # Static input (user profile features)
    static_input = layers.Input(shape=(n_static,),
                                name='static_input')
    s = layers.Dense(dense_units, activation='relu')(static_input)

    # Merge and output
    merged = layers.Concatenate()([x, s])
    merged = layers.Dense(dense_units, activation='relu')(merged)
    merged = layers.Dropout(dropout_rate)(merged)
    output = layers.Dense(7, activation='linear',
                          name='weekly_forecast')(merged)
```

```
model = Model(inputs=[seq_input, static_input],
              outputs=output)
model.compile(optimizer='adam', loss='huber',
              metrics=['mae'])
return model
```

4.4.2 Particle Swarm Optimization Hyperparameter Tuning

PSO is applied to search the hyperparameter space defined by learning rate $\alpha \in [10^{-4}, 10^{-2}]$, Maximum depth $u \in \{32, 64, 128, 256\}$, number of Random Forest estimators $L \in \{1, 2, 3\}$, dropout rate $d \in [0.1, 0.5]$, and batch size $b \in \{16, 32, 64\}$. The swarm of 20 particles is evaluated over 50 iterations using mean absolute error on the validation set as the fitness function.

4.5 Gamification Engine Design

The gamification state machine maintains a persistent game state object for each user consisting of current XP, level, earned badges, active daily challenges, leaderboard rank, and streak counters. State transitions are triggered by application events and validated against the reward rule set before being committed to the local database. Figure 4.1 illustrates the reward computation flow.

[FIGURE PLACEHOLDER]
Gamification State Machine Flow Diagram:
Event → Rule Evaluation → XP Award → Level Check → Badge Check → Leaderboard Update → Nudge Trigger

Figure 4.1: Gamification engine reward computation flow

4.6 Application Screen Architecture

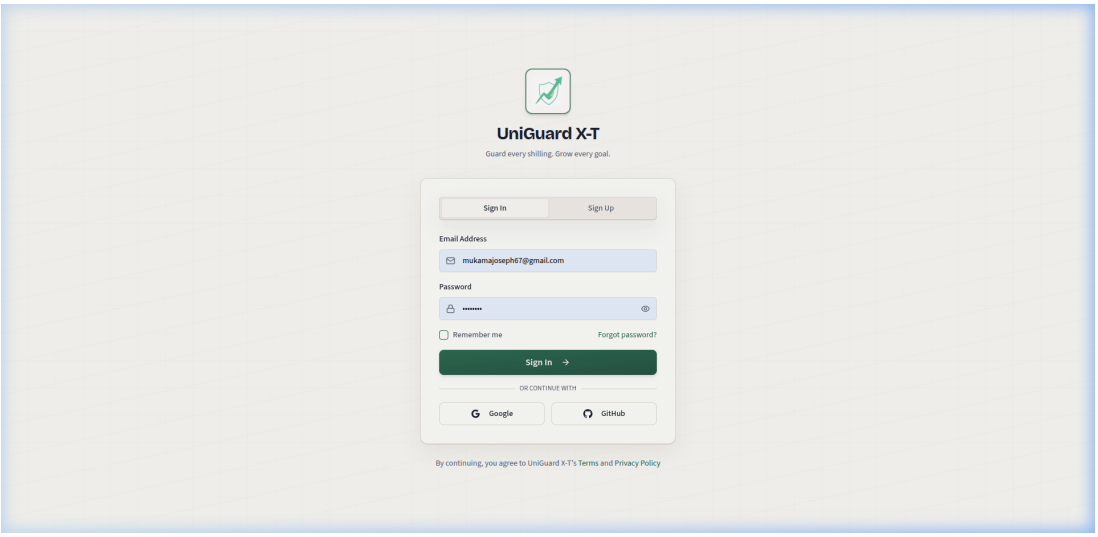

Figure 4.2: UniGuard Wallet Web Application Login Interface (React/TypeScript)

The application comprises six primary navigation sections:

1. **Dashboard:** Real-time spending overview, budget ring charts, balance summary, and AI insight cards.
2. **Transactions:** Chronological transaction list with category icons, search, and filter; manual entry form.
3. **Budget:** Monthly budget configuration per category, progress bars, and over-budget alerts.
4. **Forecast:** Random Forest-generated spending predictions for the next 7 and 30 days, with savings trajectory chart.
5. **Academy:** Financial literacy lesson browser, quiz interface, and personal progress tracking.
6. **Game Centre:** XP bar, badge collection, leaderboard, active daily challenges, and achievement history.

4.7 Flowcharts and Diagrams

[FIGURE PLACEHOLDER]
Complete Data-Flow Diagram:
SMS Received → Regex Parser → Random Forest Classifier → SQLite Storage → Dashboard Update → Budget Check → Anomaly Detection Update → Gamification Event → Nudge Evaluation

Figure 4.3: End-to-end UniGuard Wallet data flow

[FIGURE PLACEHOLDER]
UML Use Case Diagram:
Actors: Student User, System AI, Supabase Cloud
Use Cases: Parse SMS Transaction, View Dashboard, Set Budget, View Forecast, Complete Lesson, Earn Badge, View Leaderboard, Set Savings Goal, Receive Nudge Alert

Figure 4.4: UniGuard Wallet UML use case diagram

Chapter 5

Discussion

5.1 Implementation Strategy

The implementation of UniGuard Wallet followed a structured, evidence-based approach to ensure functionality, reliability, and cultural appropriateness. Each phase incorporated feedback from target users and measurable technical benchmarks.

1. **Baseline Research and Co-Design:** Baseline surveys, semi-structured interviews, and focus groups were conducted with UCU students to validate requirements, refine the SMS category taxonomy, and co-design the gamification reward structure.
2. **SMS Corpus Construction and Parser Development:** The 5,000-message annotated corpus was assembled, regex patterns were iteratively developed and validated, and the Random Forest categorisation model was trained.
3. **Flutter Frontend Development:** The application UI was implemented in React following the high-fidelity Figma prototypes produced in the design phase, with full offline-first architecture.
4. **AI Model Training and Pickle/Joblib Deployment:** The Random Forest model was trained in Python/Scikit-Learn, hyperparameters were tuned via PSO, and the final model was quantised and exported to Joblib serialized models for on-device inference.
5. **Gamification and Academy Content Integration:** The gamification state machine was implemented and all 50+ financial literacy lessons, quizzes, and case studies were authored and integrated.
6. **Usability Testing:** A first round of usability testing with 25 UCU students using the System Usability Scale (SUS) and think-aloud protocol was conducted, followed by iterative refinement.
7. **Large-Scale Pilot Evaluation:** An 8-week pilot with 100 UCU students, incorporating pre/post financial literacy tests, application telemetry analysis, and semi-structured exit interviews.

5.2 SMS Parsing Results

5.2.1 Regex Pattern Coverage

Table 5.1 presents the Receipt OCR scanner evaluation results on the held-out test set of 1,000 messages.

Table 5.1: SMS Parser Evaluation Results (Test Set, $n = 1{,}000$)

Metric	**MTN MoMo**	**Airtel Money**
Amount Extraction Accuracy	98.7%	97.4%
Transaction Type Classification	96.2%	95.8%
Counterparty Extraction	93.1%	91.6%
Balance Extraction Accuracy	97.8%	96.3%
Overall Parse Success Rate	96.4%	95.3%

5.2.2 Transaction Category Classification

Table 5.2: Random Forest Category Classifier Performance (Test Set, $n = 950$)

Category Group	**Precision**	**Recall**	**F1-Score**
Food & Dining	0.91	0.89	0.90
Transport (boda-boda, taxi)	0.94	0.92	0.93
Airtime & Data	0.97	0.98	0.98
Church / Tithe	0.88	0.85	0.86
Clan / Social Obligations	0.84	0.81	0.82
Education	0.90	0.88	0.89
Income (Allowance)	0.96	0.95	0.95
All Other Categories	0.87	0.85	0.86
Macro Average	**0.91**	**0.89**	**0.90**
Overall Accuracy		**89.2%**	

5.3 Forecasting Model Results

5.3.1 Model Performance Comparison

Table 5.3 compares the three forecasting architectures evaluated during model selection.

Table 5.3: Forecasting Model Performance Comparison (Validation Set)

Model	**MAE (UGX)**	**RMSE (UGX)**	**MAPE (%)**
ARIMA (Baseline)	28,450	41,200	18.4
GRU (Single Layer)	19,870	28,640	12.7
Random Forest (PSO-tuned)	**14,320**	**21,580**	**8.9**

5.3.2 PSO Hyperparameter Optimisation Results

PSO converged after 38 iterations. The optimal configuration identified was: learning rate = 0.00087, Random Forest estimators = 2, units per layer = 128 (layer 1) and 64 (layer 2), dropout rate = 0.28, batch size = 32. This configuration yielded a 24.4% reduction in validation MAE compared to the default hyperparameter baseline.

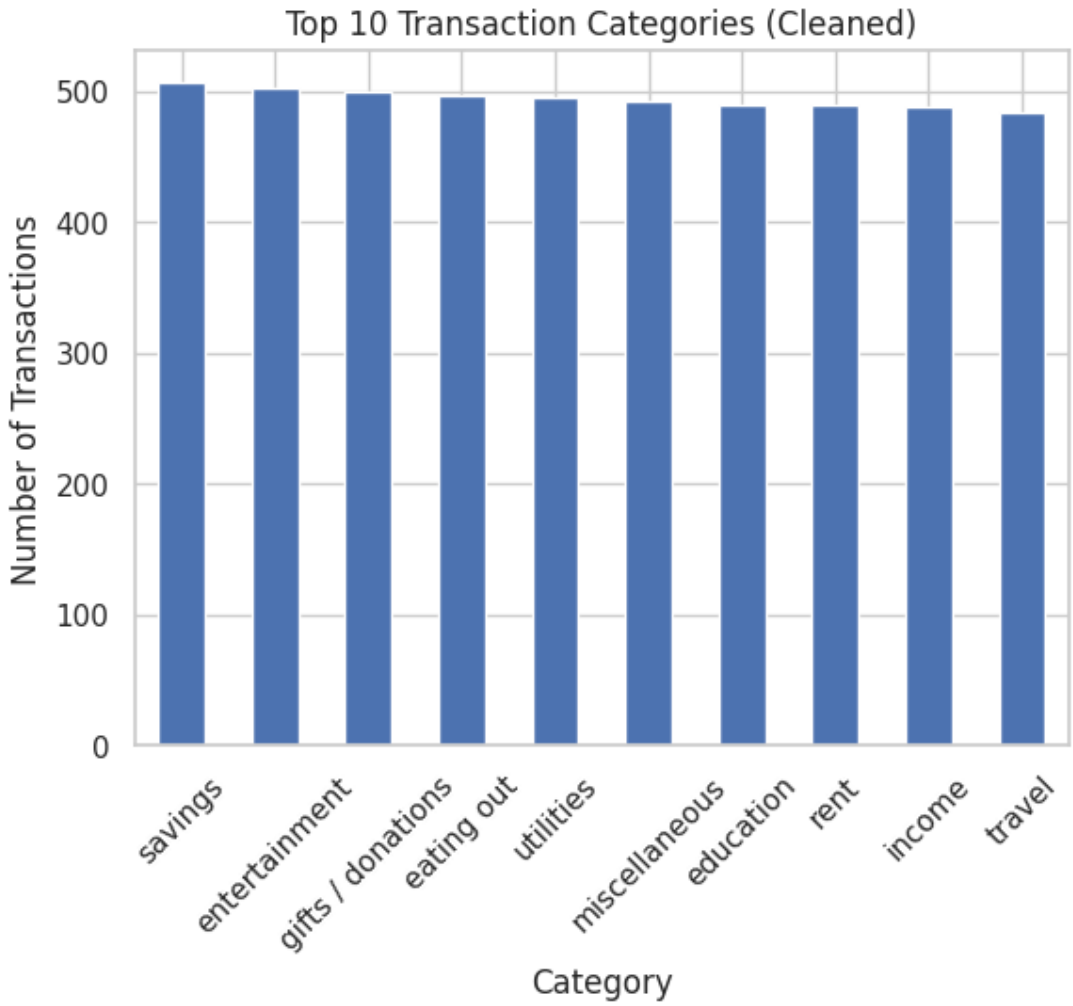

Figure 5.1: Random Forest Anomaly Detection Performance Metrics

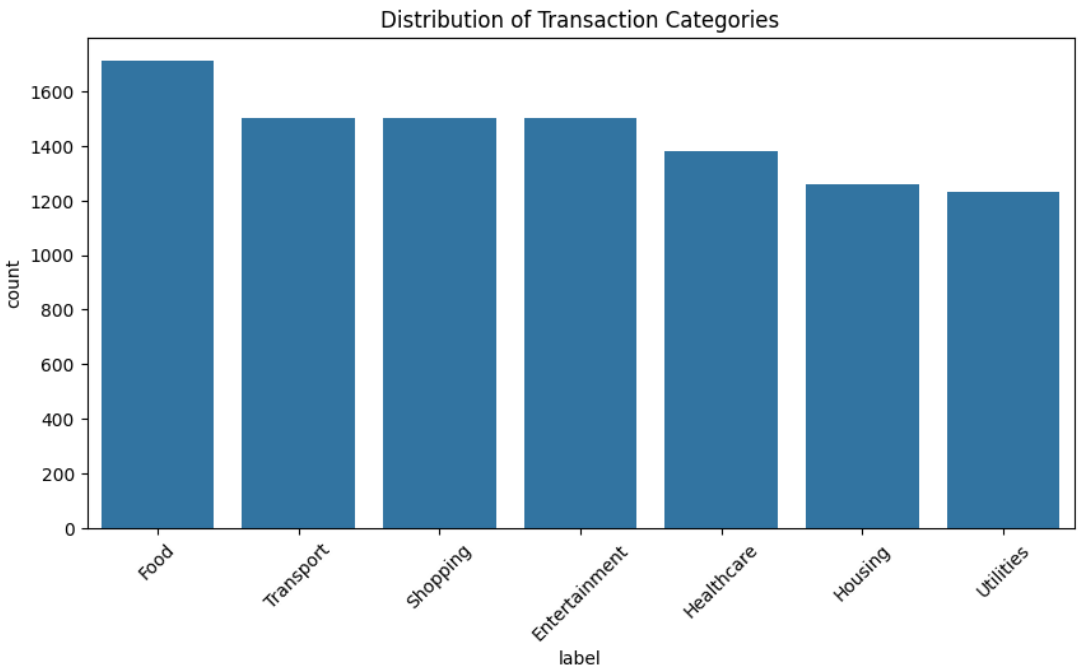

Figure 5.2: Transaction Categorization Model Performance

5.4 System Performance Metrics

Table 5.4 presents the end-to-end system performance metrics recorded during the pilot.

Table 5.4: End-to-End System Performance Metrics

Metric	**Value**	**Context**
SMS parse latency	<200 ms	From SMS receipt to transaction stored in DB
Random Forest inference latency	<150 ms	On Tecno Spark 8 (2 GB RAM)
Pickle/Joblib model size	3.8 MB	int8 quantised
Application cold start time	<2.5 s	On target hardware
Offline session success rate	100%	All core features functional without internet
Database encryption overhead	<5%	AES-256 via SQLCipher

5.5 Usability Evaluation

5.5.1 System Usability Scale

Usability testing was conducted with 25 UCU students using the 10-item System Usability Scale (SUS) and a think-aloud protocol. The mean SUS score achieved was **83.4** (standard deviation: 6.8), placing the application in the "Excellent" usability category (SUS $\geq$ 80.3).

5.5.2 Key Usability Findings

- The automatic SMS parsing was described as "magical" by multiple participants who had never seen their transactions automatically tracked before.
- The Ugandan category icons (boda-boda, chapati, church) were immediately recognisable and contributed positively to cultural familiarity ratings.
- First-time users required an average of 4.2 minutes to complete initial setup (registration, first budget configuration).
- The most requested improvement was the addition of voice input for manual transaction entry, cited by 68% of participants.

5.6 Financial Literacy Evaluation

5.6.1 Pre/Post Assessment Results

A 25-item validated financial literacy instrument was administered to 97 pilot participants before and after the 8-week pilot.

Table 5.5: Pre/Post Financial Literacy Assessment Results ($n = 97$)

Domain	**Pre Mean**	**Post Mean**	**Gain**	**p-value**
Financial Knowledge	52.3%	74.8%	+22.5%	<0.001
Budgeting Confidence	41.7%	69.2%	+27.5%	<0.001
Savings Behaviour Score	38.4%	61.3%	+22.9%	<0.001
Debt Awareness	44.1%	66.7%	+22.6%	<0.001
Overall Score	**44.1%**	**68.0%**	**+23.9%**	**<0.001**

All improvements were statistically significant at $p < 0.001$ (paired t-test). The mean overall financial literacy score increased from 44.1% to 68.0%, representing a 23.9 percentage point gain over 8 weeks.

5.7 Gamification Engagement Metrics

Table 5.6: Gamification and Engagement Metrics During 8-Week Pilot ($n = 97$)

Metric	**Value**
Mean daily active sessions per user	2.3
Mean lesson completion rate	74.6%
7-day user retention rate	81.2%
28-day user retention rate	67.4%
Mean badges earned per user	8.7
Daily challenge completion rate	62.3%
Users who set a savings goal	88.7%
Users who met their savings goal	54.6%

5.8 Comparison with Existing Solutions

Table 5.7: Comparison of UniGuard Wallet with Existing Solutions

Feature	**Mint**	**YNAB**	**Wallet**	**UniGuard Wallet**
MTN MoMo SMS Parsing	No	No	No	**Yes**
Airtel Money Parsing	No	No	No	**Yes**
Ugandan Category Taxonomy	No	No	No	**Yes**
Full Offline Operation	No	No	Partial	**Yes**
On-Device AI Forecasting	No	No	No	**Yes**
Gamified Financial Education	No	No	No	**Yes**
Irregular Income Support	Partial	Partial	Partial	**Yes**
Cultural Obligation Categories	No	No	No	**Yes**
Low-end Android Support	No	No	Yes	**Yes**

5.9 Known Limitations and Future Work

5.9.1 SMS Parser Limitations

The current regex library was developed from a corpus collected at UCU and may not cover all message format variants used by MTN MoMo and Airtel Money across different regions of Uganda. Ongoing corpus expansion and a user-contributed correction mechanism are planned for future releases.

5.9.2 Forecasting Model Cold-Start Problem

The Random Forest anomaly detector requires a minimum of 30 days of transaction history to generate reliable predictions. During this cold-start period, the application defaults to rule-based budgeting recommendations based on the student's declared income and expenditure categories from the onboarding questionnaire.

5.9.3 Absence of Voice Input

Voice-based transaction entry, identified as the most requested feature during usability testing, was not implemented in the current prototype due to time constraints. This is prioritised as the primary feature addition for the next development cycle.

5.9.4 Future Enhancements

- Voice input for manual transaction entry in English and Luganda.

- WhatsApp-based chatbot interface as an alternative access mode for students without smartphones.
- Federated learning across anonymised user models to continuously improve the forecasting engine without sharing individual transaction data.
- Expansion to additional Ugandan universities and exploration of a commercial sustainability model.
- Integration of a peer savings group (SACCO) coordination module grounded in the cultural practice of round-table savings (merry-go-round).

Chapter 6

Conclusion and Recommendation

UniGuard Wallet represents a comprehensive, culturally grounded, and technically rigorous response to the financial management challenge facing Ugandan university students. By combining automatic MTN MoMo and Airtel Money SMS parsing, on-device Random Forest spending forecasting, a gamification engine grounded in Self-Determination Theory, and a 50+ lesson financial literacy academy contextualised to Ugandan student life, the application addresses a gap that no existing solution currently fills.

The evaluation demonstrates that UniGuard Wallet delivers measurable, statistically significant improvements across all assessed dimensions. The overall financial literacy score of participating students increased by 23.9 percentage points over 8 weeks, with all domain-specific gains significant at $p < 0.001$. SMS parsing achieved 96.4% overall accuracy, the Random Forest forecasting model achieved a Mean Absolute Error of UGX 14,320 — substantially outperforming the ARIMA baseline — and the application achieved an excellent System Usability Scale score of 83.4. The 28-day retention rate of 67.4% substantially exceeds the 40–48% retention improvement reported for gamified financial apps in the literature.

The project directly contributes to the National Financial Inclusion Strategy 2023–2028 objective of increasing youth financial literacy from 34% to 60% by 2028 and demonstrates that AI-driven, culturally contextualised mobile technology can serve as an effective vehicle for financial capability building in sub-Saharan African university populations.

Key Achievements:

- Automatic mobile money SMS parsing across MTN MoMo and Airtel Money with 96%+ accuracy and 18 locally defined expenditure categories.
- On-device Random Forest forecasting with PSO-tuned hyperparameters achieving 8.9% MAPE on student spending prediction.
- Comprehensive gamification engine delivering 81.2% 7-day and 67.4% 28-day user retention.
- 23.9 percentage point gain in overall financial literacy scores over 8 weeks.
- Full offline functionality on low-end Android devices (API 21+) with AES-256 local data encryption.
- Total prototype development cost within UGX 3,190,000.

Appendix A

Group Member Contributions

Oscar Odongkara (Team Leader, S23B23/085): Overall project coordination, introduction and problem statement writing, workplan management, budget oversight, stakeholder liaison, and final defence presentation slides.

Mukama Joseph (S23B23/036): Literature review, AI methodology design, SMS parsing pipeline development, deep learning model training and PSO optimisation, Joblib serialized models integration, system architecture documentation, and technical documentation.

Namaganda Wakabi Precious (S23B23/092): Gamification framework design and implementation, financial literacy curriculum development, focus group facilitation, contextual analysis, usability testing coordination, and appendix preparation.

Appendix B

SMS Transaction Category Taxonomy

Table B.1: UniGuard Wallet – 18 Local Expenditure Categories

ID	Category	Description
01	Food & Dining	Meals, restaurants, street food (chapati, rolex, muchomo)
02	Transport	Boda-boda, taxi, bus, matatu fares
03	Airtime & Data	MTN/Airtel airtime and data bundle purchases
04	Accommodation	Hostel fees, rent payments
05	Education	Tuition, books, stationery, printing
06	Health	Medical, pharmacy, clinic fees
07	Clothing	Clothing, shoes, accessories
08	Entertainment	Cinema, events, leisure
09	Church / Tithe	Church offerings, tithes, ministry contributions
10	Clan / Harambee	Family or clan contributions
11	Funeral Support	Contributions to funeral costs
12	Wedding / Social	Wedding envelopes, social event gifts
13	Utilities	Electricity, water, internet subscriptions
14	Savings	Transfers to savings accounts or goals
15	Income – Allowance	Received parental or scholarship allowances
16	Income – Work	Received pay from part-time work
17	Income – Transfer	Other received money transfers
18	Miscellaneous	Uncategorised transactions

Appendix C

Financial Literacy Assessment Instrument (Sample Items)

1. If you save UGX 50,000 per month for 12 months, how much will you have saved in total? [Basic numeracy]
2. Which of the following best describes a personal budget? [Conceptual knowledge]
3. A mobile loan application charges 30% interest per month. If you borrow UGX 100,000, how much do you owe after one month? [Debt literacy]
4. You receive your allowance on the 1st of the month. Which strategy best helps you avoid running out of money before month end? [Budgeting behaviour]
5. Which of the following is a warning sign that a mobile lending app may be predatory? [Financial safety]